MSc Data Science

DSM150 Neural Network

Final coursework assignment

The coursework is worth 100 marks, weighted at 50% of the final mark.

Introduction

In the final coursework, you will undertake an advanced deep learning project using TensorFlow/Keras, building on the knowledge and skills developed during the midterm assignment. You will design, implement, and evaluate a deep learning model on a dataset mainly using techniques from Topic 5. However, you are expected to draw on everything you have learned throughout the course to address your chosen problem. The coursework aims to deepen your understanding of modern deep learning architectures, while developing your ability to critically analyse and communicate model performance.

You may choose the dataset introduced in Topic 5 (Cats and Dogs) or use your own dataset. If you choose to use your own unidentifiable dataset, you must be able to upload the full dataset, or—if it is too large—a smaller version of it (less than 250 MB).

Submission requirements

- Submit two files if you use the Cats and Dogs dataset:
    - The HTML export of your Jupyter Notebook.
    - The original .ipynb file.

- Submit three files if you use your own dataset
    - The HTML export of your Jupyter Notebook.
    - The original .ipynb file.
    - The dataset you used (if the dataset is larger than 250 MB then upload a smaller sample of the dataset < 250 MB)

- Your notebook must read as a report, not merely a sequence of code cells. Use clear markdown headings, subheadings, tables, and explanatory text throughout.
- You may reuse code from Deep Learning with Python (DLWP), AI, or the course video notebooks, but you must reference any code that is not your own.
- The HTML report (excludes code, tables, plots, figures and references.) must be between 2,000 and 3,000 words. Submissions exceeding this limit will not be marked.

Plagiarism reminder

Copying text, code, or analysis from other students, online sources, or generative AI tools without clear acknowledgement constitutes plagiarism and will result in disciplinary action under the University's academic integrity policy. You are expected to demonstrate your own understanding through original explanations, reasoning, and experimentation.

Background

This coursework extends the material covered in the Neural Networks module by focusing on advanced deep learning techniques, specifically deep learning for computer vision. It aligns with the module's learning outcomes by requiring you to design, train, debug, and evaluate deep learning systems using realistic datasets and contemporary methodologies. The assignment reflects real-world advanced data science practice, where model development involves iterative experimentation, justification of design decisions, and critical performance evaluation.

The assignment

For the final coursework, you will complete one substantial deep learning project using TensorFlow/Keras. You will develop a new project mainly based on Topic 5. However, you are expected to draw on everything you have learned throughout the course to address your chosen problem.
Extra credit is available for applying advanced best practices from Topic 7.

Project Structure

Your Jupyter Notebook report should include the following sections:

1. **Problem definition, dataset description and data preparation (5%)**
    - Define the task you are trying to do
    - Describe your dataset
    - Prepare your data for next steps (Note: discuss your choices)

2. **Model building, training and evaluation (20%)**
    - Build your convnet
    - Train and evaluate your model
    - Discuss the results and suggest methods to improve it

3. **Model improvement – without using pretrained models (20%)**
    - Implement your suggested method
    - Compare the new results with previous one
      (Note: Consider all the methods you learned in chapter 1 to 4 + Data augmentation)

4. **Model development – using a pretrained model with fine tuning (20%)**
    - Choose a pretrained model
    - Fine tune the model
    - Compare the results with previous section

5. **The best model (15%)**
    - Using all the methods you learned, propose your best model for your problem

6. **Visualisation of the model (10%)**
    - Visualise the filters
    - Visualise the intermediate activation
    - Visualise the heatmaps of class activation

7. **Discussion (10%)**
    - Summary of findings
    - Discuss the visualisation results

Instructions

- Work exclusively in a Jupyter Notebook.
- You must use TensorFlow/Keras.
- Credit will be awarded for effective use of standard and third-party code, with extra credit for original implementations.
- Your notebook must be clear, well-organised, and reproducible.
- You may reuse code from DLWP, AI and the video notebooks, but you must reference any code that is not your own.
- Be mindful that if you are using your own dataset, data cleaning and exploratory data analysis (EDA) should not dominate your time, as you will not receive credit for this.

How this coursework will be graded

Your coursework will be assessed based on the technical quality, clarity of explanation, and depth of your discussions when required.

Marks will reflect the appropriate and justified use of advanced deep learning techniques, including the structure of the model, methods to deal with overfitting, methods to deal with small sample size (small datasets), creative combining of all the methods and structures and interpreting the model using visualisation methods.

You will also be graded on presentation and communication, including the structure, readability, and reproducibility of your Jupyter Notebook. Proper referencing of reused code and adherence to academic integrity guidelines are mandatory.
Below you can see the standard of projects for three different categories of marks:

50–59%

- Correctly implement a convolutional neural network using TensorFlow/Keras
- Follow the required workflow (problem definition → modelling → evaluation)
- Apply at least one improvement method and one pretrained model
- Include all required sections and visualisations
- Provide basic explanations of what you did

60–69%

- Make sensible and justified modelling choices
- Apply multiple improvement techniques (e.g. regularisation, augmentation)
- Interpret training and validation results using evidence
- Clearly compare models and explain performance differences
- Write a well-structured, readable report-style notebook

70% and above

To achieve a distinction, your work should demonstrate **clear depth and insight**, including:

- Thoughtful and well-justified design decisions throughout
- Creative or effective combination of multiple techniques
- Strong evidence-based interpretation of results and visualisations
- Clear understanding of model behaviour, strengths, and limitations
- High-quality reflection and professional-level communication

At this level, your notebook should read like a **mini research report**, showing synthesis rather than isolated experiments.

References

Chollet, F. *Deep learning with Python*. (Manning Publications, 2017) first edition. Chapter 1 – 8.

[END OF COURSEWORK ASSIGNMENT]